HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F72ZA1
FINANCIAL MATHEMATICS 1

4 December 2006

Time allowed 2 hours

Attempt all **7** questions

Total Marks 60

Only approved calculators may be used.

NUMERICAL SOLUTIONS

1. Fill in the missing values in the following table:

i	v	d	$i^{(4)}$	$d^{(2)}$
0.111111	0.900000	0.10	0.106760	0.102633

2. (a) £103.12
   (b) £141.35

3. (a) Bookwork proof
   (b) (i) 7.5152
   (ii) 2.8111
   (iii) 34.7391

4. $100,000 = \frac{X}{5}(Ia)_{\overline{5}|}^{0.06} + X(1.06)^{-5}a_{\overline{5}|}^{0.06} + X(1.06)^{-10}a_{\overline{10}|}^{0.07}$.
   Hence $X = £10,527.38$.

5. (a) (i) £12,214.34
   (ii) £152,217.67
   (iii) £7,610.88
   (b) £67,466.31

6. (a) £83.82 (note CGT is payable)
   (b) 11.8% *per annum*

7. (a) The yield in 2006–7 is $i_1 = 0.014953$. The yield in 2008–9 is $i_2 = 0.017303$. Hence LIRR is given as the solution of:

$$(1 + \text{LIRR})^4 = (1 + i_1)^2(1 + i_2)^2 \Rightarrow \text{LIRR} = 0.0161.$$

   (b) TWRR is given by:

$$\left(\frac{1.1}{1.12}.\frac{0.9}{0.85}.\frac{0.85}{0.8}.\frac{0.95}{0.97}\right)^{1/4} - 1 = 0.0199$$